Car Rental Project

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Project’s Composition

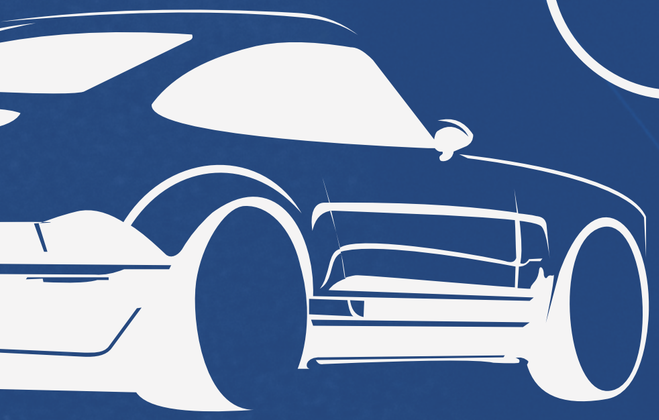
Introduction

ER Diagram

Relational Schema

SQL Declaration

SQL Statements



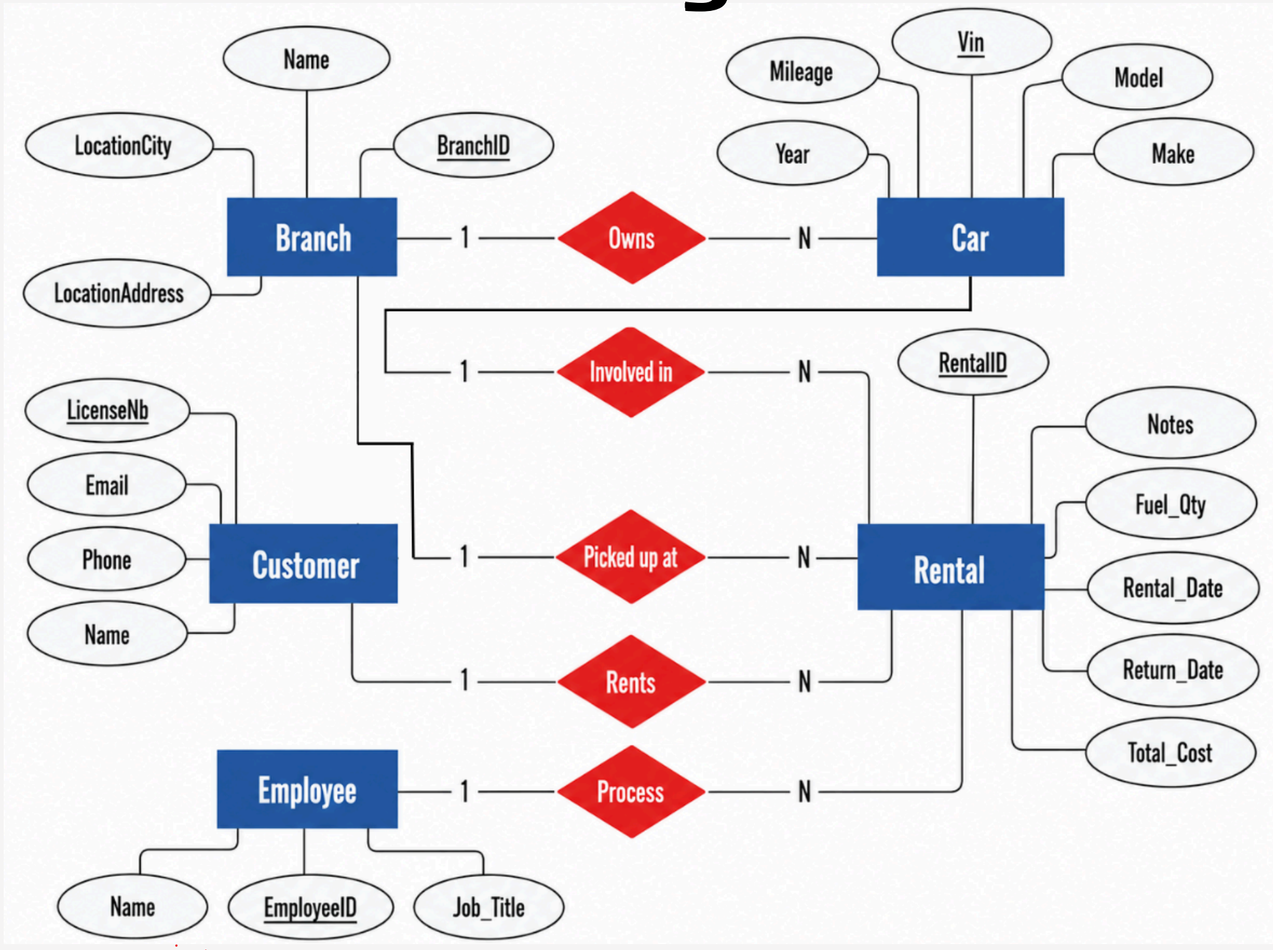
Introduction

A nationwide car rental agency needs a database to manage its fleet and customer rentals. Each agency has a unique identifier, a name, and an address. Cars, identified by a VIN, are assigned to an "origin" agency. Customers, with a unique driver's license number, provide their name, phone number, and email. Each rental records the start date, expected return date, total cost, fuel quantity, and any potential damage. Each transaction is managed by an employee with a unique ID and a job title.





ER Diagram


Name
LocationCity
BranchID
LocationAddress
Branch
1
Owns
N
Car
Vin
Mileage
Model
Year
Make
1
Involved in
N
RentalID
LicenseNb
Email
Phone
Name
Customer
1
Picked up at
N
Rental
Notes
Fuel_Qty
Rental_Date
Return_Date
Total_Cost
1
Rents
N
Employee
1
Process
N
Name
EmployeeID
Job_Title

Relationship Summary

- Branch OWNS Car: One Branch owns many Cars.
- Rental PICKED UP IN Branch: One Branch is involved in many Rentals.
- Customer RENTS Rental: One Customer rents many Rentals.
- Employee PROCESSES Rental: One Employee processes many Rentals.
- Car INVOLVED IN Rental: One Car can be used in many Rentals



Relational Schema

- Branch(BranchID, Name, LocationCity, LocationAddress)
- Car(Vin, Make, Model, Year, Mileage, BranchID)
- Customer (LicenseNb, Name, Email, Phone)
- Employee (EmployeeID, Name, Job_Title)
- Rental (RentalID, Rental_Date, Return_Date, Fuel_Qty,Total_Cost, Notes, Vin#, LicenseNb#, BranchID#, EmployeeID#)





SQL Declaration

```
CREATE TABLE "Branch" (
 "BranchID" INTEGER NOT NULL UNIQUE,
 "Name" TEXT NOT NULL UNIQUE,
 "LocationCity" TEXT NOT NULL,
 "LocationAddress" TEXT NOT NULL UNIQUE,
 PRIMARY KEY("BranchID")
);
CREATE TABLE "Customer" (
 "LicenseNb" INTEGER NOT NULL UNIQUE,
 "Name" TEXT NOT NULL,
 "Email" TEXT UNIQUE,
 "Phone" TEXT NOT NULL UNIQUE,
 PRIMARY KEY("LicenseNb")
);
CREATE TABLE "Employee" (
 "EmployeeID" INTEGER NOT NULL UNIQUE,
 "Name" TEXT NOT NULL,
 "Job_Title" TEXT NOT NULL,
 PRIMARY KEY("EmployeeID")
);
CREATE TABLE "Car" (
 "Vin" TEXT NOT NULL UNIQUE,
 "Make" TEXT NOT NULL,
 "Model" TEXT NOT NULL,
 "Year" INTEGER NOT NULL,
 "Mileage" INTEGER NOT NULL,
 "BranchID" INTEGER NOT NULL UNIQUE,
 PRIMARY KEY("Vin"),
 FOREIGN KEY("BranchID") REFERENCES "Branch"
("BranchID")
);
```

```
CREATE TABLE "Rental" (
 "RentalID" INTEGER NOT NULL UNIQUE,
 "Rental_Date" TEXT NOT NULL,
 "Return_Date" TEXT NOT NULL,
 "Fuel_Qty" REAL NOT NULL,
 "Total_Cost" REAL NOT NULL,
 "Notes" TEXT,
 "Vin" TEXT NOT NULL UNIQUE,
 "LIcenseNb" INTEGER NOT NULL UNIQUE,
 "BranchID" INTEGER NOT NULL UNIQUE,
 "EmployeeID" INTEGER NOT NULL UNIQUE,
 PRIMARY KEY("RentalID"),
 FOREIGN KEY("BranchID") REFERENCES "Branch"
("BranchID"),
 FOREIGN KEY("EmployeeID") REFERENCES "Employee"
("EmployeeID"),
 FOREIGN KEY("LIcenseNb") REFERENCES "Customer"
("LicenseNb"),
 FOREIGN KEY("Vin") REFERENCES "Car"("Vin")
);
```

Some SQL Statements

1. High-Mileage Vehicles (Most Used First)

```
SELECT Vin, Make, Model, Mileage
FROM Car WHERE Mileage > 30000
ORDER BY Mileage DESC;
```

2. Low-Fuel Returns

```
SELECT RentalID, Fuel_Qty, Return_Date
FROM Rental WHERE Fuel_Qty < 0.25
ORDER BY Fuel_Qty ASC;
```

3. Customer "Blacklist" Check

```
SELECT Name, LicenseNb
FROM Customer
WHERE Email IS NULL OR Phone IS NULL;
```

4. Busy Employees

```
SELECT EmployeeID,
COUNT(RentalID) AS Total_Processed
FROM Rental
GROUP BY EmployeeID
ORDER BY Total_Processed DESC;
```



Thank You

For Your Attention